

# Milestone 2: Data Preparation/Collection & Cleaning 149/150 Points

3/6/2026

Attempt 5



Review Feedback

3/6/2026

Attempt 5 Score:

149/150



View Feedback

Anonymous Grading: no

Unlimited Attempts Allowed

3/6/2026

▼ Details

## Overview

This project is a **comprehensive group assignment** that allows you to apply concepts learned in this course. You will work with **real data** to answer research questions, perform **data mining techniques**, and communicate findings through a **professional project website**.

The project consists of **four milestones** to ensure structured progress. Each milestone builds upon the previous one, and missing any prior requirement must be addressed before proceeding.

**Project Format:** Hosted on a **GitHub repository** and a **public website**

**Group Work:** Teams of **three students**

**Final Deliverable:** **Website + Report (8–12 pages in ACM format) + Source Code**

### Submission Guidelines

All project materials should be submitted via Canvas in the following format:

- **Website URL** (must be publicly accessible)
- **GitHub Repository URL** (including datasets, code, and reports)
- **PDF Report** (full project documentation in ACM format)
- **ZIP File** (containing source code and key results)
- **Team Contribution Report** (detailing individual contributions)

## Milestone 2: Data Collection & Cleaning

### Objective

In this milestone, students will **collect, clean, and explore their datasets** while ensuring **high data quality and integrity**. This step is crucial in preparing the data for analysis and modeling.

## Requirements

Create a **"Data Exploration" Tab** on your website which showcases the following:

### Data Collection

Data should be collected using **dynamic or authoritative sources** whenever possible. Acceptable sources include:

- **APIs**  
Any relevant API that provides live or periodically updated data.

- **Cloud-hosted datasets**

Platforms such as Google BigQuery, AWS Open Data, or Azure Open Datasets.

- **Streaming or real-time data**

Examples include IoT feeds, stock prices, weather data, or Google Trends.

- **Government or open data portals**

Sources such as Data.gov, World Bank, WHO, UN, or similar official repositories.

- **Crowdsourced or sensor data**

For example, OpenStreetMap, Fitbit/Strava exports, surveys, or user-generated data.

- **Web scraping**

Allowed **only if** no reliable API or hosted dataset exists. The choice must be clearly justified.

If an API or dynamic source is **not available**, students **may directly download data from an official or authoritative source**, provided they clearly explain:

- why an API or live data source was not feasible, and
- how the data is still relevant, reliable, and appropriate for the research questions.

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## Documentation Requirements

Students must:

- Clearly describe the **data source** and **method of collection**
- Justify why the dataset is **relevant to the research questions**
- Provide **before-and-after snapshots** of the dataset (raw vs processed)

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## Originality Requirement

The project must demonstrate **original data collection or thoughtful sourcing**.

- Purely downloading **pre-packaged, static datasets from platforms like Kaggle without justification** is discouraged.
- Priority should be given to **APIs, cloud-hosted datasets, streaming data, or official sources**.
- Direct downloads are acceptable **only when APIs or dynamic access are unavailable and the choice is well-justified**.

Failure to follow these guidelines may result in penalties for this milestone.

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## Data Cleaning and Preprocessing

Each team must document how they prepared their dataset for analysis. Include:

- **Handling Issues/Noise in the Data:**

Missing values (imputation, removal, etc.), duplicates and inconsistencies, outliers and anomalies.

- **Understanding the Data:**

Identify data types (categorical, numerical, text, etc.).

Apply transformations (normalization, standardization, log transformations).

Perform dimensionality reduction if necessary (e.g., PCA).

- **Basic Statistical Analysis:**  
Compute **summary statistics** (mean, median, variance, skewness, etc.).  
Explore **correlations** between features.  
Conduct **data similarity and integration analysis** if applicable.
- **Data Quality Assessment:**  
Evaluate the dataset for **completeness, consistency, and usability**.  
Ensure the dataset aligns with the **project's research objectives**.  
(Teams should also include a short reflection on **data ethics, bias, or limitations** relevant to their dataset.)
- **Advanced Data Understanding:**  
Perform visual diagnostics like **QQ plots** to assess normality.  
Examine relationships and consolidate datasets (if applicable).  
Normalize and transform data for modeling.

Visualizations

- Create at least **ten unique and meaningful visualizations**.
- Provide **brief descriptions and insights** for each visualization.
- Avoid redundant charts (e.g., **10 histograms will result in a penalty**).
- To add variety, students may also use tools such as **Tableau, Power BI, or Looker** in addition to Python visualizations (optional).

Before/After

- Show dataset snippets **before and after cleaning**.
- Highlight **major transformations** and improvements in data quality.

Submission Requirements

- **Website URL** (updated with the "Data Exploration" tab)
- **GitHub URL** (including dataset and Python code; include a **requirements.txt** file for reproducibility)
- **PDF Report** (cover page, abstract, data acquisition, preprocessing, and visualization summaries)

Reminder: All work must be original. **Plagiarism or uncredited work will be penalized** according to university policy.

View Rubric

Select Grader

Taihei Sone (TA) ▾

Some Rubric (4)

| Criteria                                                              | Ratings              |                   | Pts         |
|-----------------------------------------------------------------------|----------------------|-------------------|-------------|
| Introduction and Teams Tab<br><a href="#">view longer description</a> | 10 pts<br>Full Marks | 0 pts<br>No Marks | 10 / 10 pts |

Some Rubric (4)

| Criteria                                                      | Ratings              |                   | Pts               |
|---------------------------------------------------------------|----------------------|-------------------|-------------------|
| Data Collection<br><a href="#">view longer description</a>    | 40 pts<br>Full Marks | 0 pts<br>No Marks | 40 / 40 pts       |
| Data Cleaning<br><a href="#">view longer description</a>      | 40 pts<br>Full Marks | 0 pts<br>No Marks | 40 / 40 pts       |
| Data Visualisation<br><a href="#">view longer description</a> | 30 pts<br>Full Marks | 0 pts<br>No Marks | 30 / 30 pts       |
| Report<br><a href="#">view longer description</a>             | 20 pts<br>Full Marks | 0 pts<br>No Marks | 20 / 20 pts       |
| Submissions<br><a href="#">view longer description</a>        | 10 pts<br>Full Marks | 0 pts<br>No Marks | 9 / 10 pts        |
|                                                               |                      |                   | Total Points: 149 |

Website: <https://aazodpe.github.io/> (<https://aazodpe.github.io/>)

Github: <https://github.com/aazodpe/aazodpe.github.io> (<https://github.com/aazodpe/aazodpe.github.io>)



(<https://canvas.colorado.edu/courses/133037/modules/items/7262787>)



(<https://canvas.colorado.edu/courses/133037/mo>)